

Ground Rules

General:

- Your learning is your job. We can support but not take responsibility for your learning.
- Utilize your mentors time wisely. Be clear on what you need before approaching mentor. Send it in writing (email or chat) so mentor can come prepared.
- If you do not have a task assigned, Contact your mentor immediately.
- If you are not able to complete task on time. Inform mentor before mentor asks you.
- Everyone must log/report progress by end of day with following details
 - a. What did I do today?
 - b. what will I do tomorrow?
 - c. Do I have any blockers?

@DSR should have 1 or 2 lines describing the outcome of your work. Not the process.
For example

Wrong: I will watch videos about Asp.Net, I will practice asp.net and then try to complete the Project Page task.

Correct: I will complete Project Page Task

Coding

- Everyone must use Prolifics GitLab. @Sundeep to confirm if it makes sense to create separate tasks in Prolifics GitLab.
- If you have specific question, post your question on Stack Overflow before approaching mentor. Send SO link

Resources

- Code Quality: [Sonar Lint extension](#)
- Coding Standards: [C# Coding Standards Best Practices - Dofactory](#)
- Unit Testing /NUnit:
- VS Code

Project Details

1. **Create:** Create an Application by name ProlificsProjectManager(PPM in short)

Create program to support following operations. Every time Add operation is performed. Store it in respective Array

- 1) Add Project
- 2) View Projects
- 3) Add Employee
- 4) View Employees
- 5) Add Role
- 6) View Roles
- 7) Quit

2. **Enhance:** Enhance above program with following features:

- 2.1. Create additional C# projects to the solution. Namely PPM.Model , PPM.Domain, PPM.UI.Console , PPM.Main
- 2.2. Add additional options for user. Your revise Options/Menu will be as follows. **Note:** Each Employee will have a Role. Project will have zero or more Employees.
 - 1) Add Project - Id, Name, Start date, End date
 - 2) View Projects
 - 3) Add Employee - Id, First name, Last name, Email, Mobile, Address, RoleId
 - 4) View Employees
 - 5) Add Role - Id, Name
 - 6) View Roles
 - New Options start --
 - 7) Add Employee to project
 - 8) Delete Employee from project
 - 9) View Project Detail (Project Details and List of employees by role)
 - New Options End --
 - 10) Quit
3. **Code Quality:** Modify your program to make it higher quality as follows
 - 3.1. **Coding standards:** Read about coding standards. This [Coding standards link](#) is a good start. Modify your code to follow coding standards described here.
 - 3.2. **Code Quality Check:** Install [SonarLint visual studio extension](#) and make sure warnings reported by Sonar tool are resolved.
 - 3.3. **Unit Tests:** Add NUnit Tests for the above tasks. @Raghu will help with NUnit reading material and can guide if needed
4. **Modularize:** Modify above program to modularize it.
 - 4.1. Ask Input by module. Revised Menu looks as follows. Note, that we will have 2 steps before reaching operation. First is Module and second is Operation on that Module
 1. Project Module
 - 1.1. Add
 - 1.2. List All
 - 1.3. List By Id
 - 1.4. Delete
 - 1.5. Return to Main Menu
 2. Employee Module
 - 2.1. Add
 - 2.2. List All
 - 2.3. List By Id
 - 2.4. Delete
 - 2.5. Return to Main Menu
 3. Role Module
 - 3.1. Add

- 3.2. List All
 - 3.3. List By Id
 - 3.4. Delete
 - 3.5. Return to Main Menu
 - 4. Quit
 - 4.2. While adding a project, also provide option to Add an Employee with role.
 - 4.3. Validation: If Employee or Role is deleted, must check whether Employee or role is mapped to any project.
 - 4.4. Interfaces: Modify operations above to implement IEntityOperation interface
5. **Save State:** Modify above program to Save App state.

Add additional option to save the App data into an XML file. Serialize and save this.

- 1. Project Module
 - a. Add
 - b. List All
 - c. List By Id
 - d. Delete
 - e. Return to Main Menu
 - 2. Employee Module
 - a. Add
 - b. List All
 - c. List By Id
 - d. Delete
 - e. Return to Main Menu
 - 3. Role Module
 - a. Add
 - b. List All
 - c. List By Id
 - d. Delete
 - e. Return to Main Menu
 - 4. Save
 - 5. Quit
6. **Additional Save Options:** Modify above program to add additional options and Track Timestamp
- I. Add the database tables to store data with below content and add additional database fields to store the Timestamp tracking for Creation Date "CreatedOn" & Modification Date "ModifiedOn"
 - II. Create a DAL with ADO.NET connectivity and perform all these CRUD operations in the database
 - 1. Project Module
 - a. Add
 - b. List All

- c. List By Id
 - d. Delete
 - e. Return to Main Menu
- 2. Employee Module
 - a. Add
 - b. List All
 - c. List By Id
 - d. Delete
 - e. Return to Main Menu
- 3. Role Module
 - a. Add
 - b. List All
 - c. List By Id
 - d. Delete
 - e. Return to Main Menu
- 4. Quit

III. Create a DAL with Entity framework and perform all these CRUD operations in the database

- 1. Project Module
 - a. Add
 - b. List All
 - c. List By Id
 - d. Delete
 - e. Return to Main Menu
- 2. Employee Module
 - f. Add
 - g. List All
 - h. List By Id
 - i. Delete
 - j. Return to Main Menu
- 3. Role Module
 - k. Add
 - l. List All
 - m. List By Id
 - n. Delete
 - o. Return to Main Menu
- 4. Quit

7. **API Project:** Create API project.

6.1) Create another csharp project called PPM.API and reference/utilize existing functionality to provide API

8. Web Project: Create Web Project

7.1) Create another csharp project called PPM.UI.Web

7.2) Provide above operations as buttons in Web UI

7.3) Login Should be created and User & Password should be used from the Employee Table

Employee-Email : Username

Employee-Password : Password (Add new field)

7.4) Add Roles Admin, Manager, User

7.5) User logged as Admin can only create Employees

7.6) User logged as Manager can only create Projects & Map Employees

7.6) User logged as Role User can only view Projects List assigned to him

7.7) All User can view their own Profiles

9. Redo the same tasks using Java. Use Hibernate instead of

Core Tasks, For Reference:

1. Create a console app to store and retrieve in-memory list of employees and projects

2. Adding object dependency in model

3. Creating class library for model and in-memory list object

4. Creating interfaces and introduce dependency injection

5. Create file-based object storage

6. Create the above project using Java

7. Create API (.Net and spring boot) that can be called from console app

8. Create ADO.Net/EF Core and hibernate

9. Create presentation layer using HTML, CSS, and JS